

Semester Two of Academic Year (2014---2015) of BJUT

《 University Physics 》 Exam Paper

Exam Instructions : Answer ALL Questions

Honesty Pledge :

I have read and clearly understand the Examination Rules of Beijing University of Technology and am aware of the Punishment for Violating the Rules of Beijing University of Technology. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I would accept the punishment thereof.

Pledger : _____

Class No : _____

BJUT Student ID : _____

UCD Student ID _____

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Notes:

The exam paper has 3 parts on 6 pages, with a full score of 100 points. You are required to use the given scratch paper attached only.

Total Score of the Exam Paper (For teachers' use only)

Item	Multiple Choice	Blank Filling	Calculation					Total Score
Full Score	30	20	10	10	10	10	10	
Obtained Score								

Obtained score

Part 1: Multiple Choice (Each answer is worth 3 points, total score is 30)

1. A mass point executes curvilinear movement on a plane. Its instantaneous velocity is $\vec{v}$, instantaneous speed is v , average speed is $\bar{v}$, and average velocity is $\bar{\vec{v}}$. Which of the following relationships is right?

- (A) $|\vec{v}| = v, |\bar{\vec{v}}| \neq \bar{v}$ (B) $|\vec{v}| \neq v, |\bar{\vec{v}}| \neq \bar{v}$
 (C) $|\vec{v}| = v, |\bar{\vec{v}}| = \bar{v}$ (D) $|\vec{v}| \neq v, |\bar{\vec{v}}| = \bar{v}$ []

2. An object moves along a straight line from v_0 with $\frac{dv}{dt} = -kvt$ (k is a constant). What is the function of $v(t)$?

- (A) $v = v_0 e^{\frac{1}{2}kt^2}$ (B) $v = v_0 e^{kt^2}$
 (C) $v = v_0 e^{-\frac{1}{2}kt^2}$ (D) $v = v_0 e^{-kt^2}$ []

3. A variable force acts on an object from $t_i = 0$ to t_f . The impulse of the force is zero. One can conclude that

- (A) $\Delta \vec{r} = 0$ and $\Delta \vec{p} = 0$.
 (B) $\Delta \vec{r} = 0$ but possibly $\Delta \vec{p} \neq 0$.
 (C) possibly $\Delta \vec{r} \neq 0$ but $\Delta \vec{p} = 0$.
 (D) possibly $\Delta \vec{r} \neq 0$ and possibly $\Delta \vec{p} \neq 0$. []

4. Which one of the following statements is right?

- (A) If the impulse of the external force is zero, the work done by this force is definitely zero.
 (B) If the kinetic energy doesn't change, its momentum doesn't change.
 (C) If the momentum of the mass point changes, the kinetic energy definitely changes.
 (D) If the work done by the external force is zero, the impulse of the external force is definitely zero. []

5. The change in mechanical energy of a closed system is equal to the work done on it by

- (A) all forces acting in the system.
 (B) all conservative forces acting in the system.
 (C) all non-conservative forces acting in the system.
 (D) all the external forces and the internal non-conservative forces acting in the system []

6. A physic professor is sitting on a rotating chair with her arms outstretched, each holding a medium sized barbell. The frictionless chair is originally rotating at a constant angular speed. She then pulls her arms closer her body. When she brings her arms in, her angular momentum

- (A) increases.
- (B) remains constant.
- (C) decreases.
- (D) changes, but weather it increases or decreases depends on how she brings her arms in.

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7. A particle on a spring executes simple harmonic motion. When the particle is found at

$x = x_{\max}/2$ the speed of the particle is

- (A) $v_x = v_{\max}$.
- (B) $v_x = \sqrt{3}v_{\max}/2$.
- (C) $v_x = \sqrt{2}v_{\max}/2$.
- (D) $v_x = v_{\max}/2$.

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8. There is a standing wave in a string and it is a node at point $x=0$. The incident wave function is

$y_1 = 2 \times 10^{-2} \cos(2\pi t - 20\pi x + \frac{\pi}{4})\text{m}$. Then the reflected wave function is

- (A) $y_2 = 2 \times 10^{-2} \cos(2\pi t - 20\pi x - \frac{\pi}{4})\text{m}$
- (B) $y_2 = 2 \times 10^{-2} \cos(2\pi t + 20\pi x + \frac{\pi}{4})\text{m}$
- (C) $y_2 = 2 \times 10^{-2} \cos(2\pi t + 20\pi x + \frac{3\pi}{4})\text{m}$
- (D) $y_2 = 2 \times 10^{-2} \cos(2\pi t + 20\pi x - \frac{3\pi}{4})\text{m}$

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9. Two events A and B are simultaneous but occur at different locations in frame S . In any other frame S' moving relative to frame S at a constant velocity in the direction of positive x ,

- (A) the events could be simultaneous and occur at the same location.
- (B) the events could either be simultaneous or occur at the same location, but not both.
- (C) the events could be both simultaneous and at the same location.
- (D) the events will be neither simultaneous nor at the same location.

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10. The root-mean-square speed of molecules in still air at room temperature is closest to

- (A) walking speed(2m/s).
- (B) the speed of a fast car(30m/s).
- (C) the speed of a supersonic airplane(500m/s).
- (D) escape speed from Earth(1.1×10^4 m/s)

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Obtained score

Part 2: Blank Filling (total score is 20)

11. (5 points) A mass point moves on a circle of radius $R = 2\text{m}$. The speed is zero initially. The tangential acceleration is $a_t = 2\text{m/s}^2$. Then the magnitude of centripetal acceleration a_n is _____ at $t = 1\text{s}$. The moving distance of the mass point is _____ from $t = 0$ to $t = 1\text{s}$.

12. (3 points) An object with mass m is projected out from point O with initial velocity v_0 , the angle between v_0 and the horizontal plane is α and we ignore the air resistance. The impulse of the gravity of the object moving from point O to its landing point in the same horizontal plane is _____.

13. (3 points) Suppose a particle takes part in three simple harmonic motions of the same frequency in the same direction respectively expressed by $x_1 = A\cos(\omega t + \pi/3)$, $x_2 = A\cos(\omega t + 5\pi/3)$ and $x_3 = A\cos(\omega t + \pi)$, the equation of the resultant motion is _____.

14. (5 points) Observer A moves with a speed of $v = 0.8c$ relative to observer B. Observer A carries an object with $m = 1\text{kg}$ with himself. The total energy measured by observer A is _____, and the total energy measured by observer B is _____.

15. (4 points) A quantity of gas containing N molecules has a speed distribution function $f(v)$.

The number of molecules with speeds between v_1 and v_2 at an equilibrium state is

_____. The average speed $\bar{v}$ at equilibrium state is

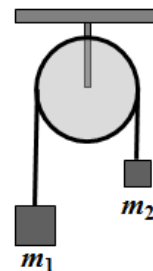
_____.

Obtained score

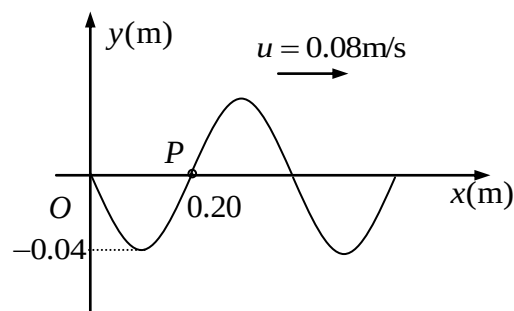
Part 3: Calculation (Each answer is worth 10 points, total score is 50)

16. A meteorolite m falls from height h_1 to h_2 . If the air resistance is negligible, find the work done on the meteorolite by the gravity. (R and M are radius and mass of the Earth and G is gravitational constant)

17. A pulley can be considered as a uniform disk of mass M and radius R , mounted on a fixed (frictionless) horizontal axle. Two blocks of mass m_1 and m_2 ($m_1 > m_2$) hang from a light cord that crosses the rim of the pulley (Suppose the cord does not slip or stretch). If the system is at rest in the beginning, calculate the angular speed of the pulley at time t . (The moment of inertia of the pulley is $J = \frac{1}{2}MR^2$)



18. The wave shape curve for a simple harmonic wave at $t = 0$ is shown in figure. Calculate (a) the wave equation; (b) the oscillation equation of point P .



19. A pion is an unstable particle with an average lifetime of 2.60×10^{-8} s (measured in the rest frame of the pion). (a) If the pion is made to travel at very high speed relative to a laboratory, its average lifetime is measured in the laboratory to be 4.20×10^{-7} s. Calculate the speed of the pion expressed as a fraction of c . (b) What distance, measured in the laboratory, does the pion travel during its average lifetime?

20. The pV -diagram shows a cycle of a heat engine that uses 0.5 mole of an ideal gas having $\gamma = 1.40$. The curved part ab of the cycle is adiabatic. (a) Find the pressure of the gas at point a . (b) How much heat enters this gas per cycle, and where does it happen? (c) How much heat leaves this gas in a cycle, and where does it occur? (d) How much work does this engine do in a cycle? (e) What is the thermal efficiency of the engine?

